

Draftly - One Pager

Selected Topic: Draftly – Gmail AI Reply Agent

Objective

Draftly addresses the operational overhead of routine professional communications. By automating email confirmations and follow-ups, the system improves productivity while ensuring stylistic consistency. The objective is to build a reliable, extensible Gmail AI agent that fetches incoming emails, generates high-context drafts, learns user style, and provides a strict review-before-send workflow.

Step-by-Step Approach

1. Secure Integration & Data Ingestion * **OAuth2 Authorization:** Implement a secure OAuth2 flow for users to authenticate and connect their Gmail accounts. Encrypt and store tokens securely using AES-256. * **Email Syncing:** Develop scheduled background workers to periodically fetch incoming emails and their metadata (Sender, Subject, Thread ID) via the Gmail API, strictly adhering to rate limits and quotas.

2. Intelligent Triage & Contextual Profiling * **Triage Pipeline:** Process incoming threads through an AI classifier to evaluate urgency and determine if a reply is needed (`reply_needed`). * **Persona Learning:** Analyze the user's recently sent emails to build a personalized communication profile, capturing their preferred tone, greeting, and closing styles.

3. Automated Draft Generation * **Asynchronous Processing:** Utilize a robust message broker (e.g., Redis + BullMQ/Celery) to orchestrate draft generation without blocking the main API thread. * **LLM Integration:** Feed the thread history and the user's learned persona into a Large Language Model (LLM) to generate highly contextual, stylized, and accurate reply drafts.

4. Review, Approval & State Management * **API Layer:** Expose RESTful APIs for the frontend to retrieve threads, triage statuses, and the generated drafts. * **Workflow States:** Implement a strict state machine for drafts (Generated → Edited → Approved / Rejected). This ensures an "Approved-only" dispatch logic, preventing rogue automated emails.

5. Native Synchronization & Reliable Dispatch * **Gmail Draft Sync:** Automatically mirror the AI-generated drafts to the user's actual Gmail "Drafts" folder for seamless

cross-platform accessibility. * **Thread Integrity:** Upon user approval, dispatch the email via the Gmail API, carefully injecting In-Reply-To and References headers to maintain native email threading. * **Failure Handling:** Implement idempotency keys for the sending logic and automatic retry mechanisms for network or token expiration failures.

6. Persistence & Security * **Database:** Use a relational database (PostgreSQL) to persist user profiles, email metadata, draft states, and historical logs. * **Security Guardrails:** Enforce strict access control (Auth/Authz) on all endpoints and encrypt sensitive user preferences to guarantee data privacy.